

2080

Bachelor of Science in Computer Science and Information Technology

Course Title: **Computer Networks**

Code No: **CSC263** Semester: **4th Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. What is classful addressing? Explain link state routing with a suitable example.
2. What is flow control? How does stop-and-wait ARQ protocol handle error? What are its disadvantages?
3. Explain TCP header with a neat diagram. Highlight on its uses.

Section B

Attempt any Eight questions[8×5=40]

4. Explain different types of network topologies.
5. What do you understand by circuit switching? Explain. What are its advantages and disadvantages?
6. State the functionality of logical link control. Briefly explain HDLC.
7. Differentiate between unicast and multicast routing.
8. What is the difference between port and socket? Explain leaky bucket algorithm with an example.
9. What do you understand by DNS? Explain FTP and SFTP.
10. Explain the architecture of SDN.
11. Suppose you are given an IP address 192.168.0.0, perform subnetting and divide the given network into 2 subnets. Calculate total number of hosts that can be configured, range of the IP addresses.
12. Write short notes on:
a. IPv4
b. Infrared